URP 全套英文图逐图解析与源表索引

本索引按研究阶段收录可取得的正式英文图及诊断图。每图包含英文图注、中文分析、适用边界和源表。旧文件 F/R/S 编号用于追溯，最终论文重新编号为 Figure 1—14。R02 已按用户确认的原 CI 图解释；四年回归和新前沿另有本轮修订。

1. F02_CC_trajectories_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Figure 2. Simulated daily canopy cover (solid lines) and calibration data (open circles) for (a) 2019, (b) 2020, (c) 2021 and (d) 2022. Days after planting follow the model DAP field. R^2cal = 1 − SSE/SST and RMSE are calculated at paired calibration dates; these are in-sample assessments. Lines are trajectories at frozen parameter centers, not posterior predictive means or uncertainty intervals.

解析：共 48 对；年度校准 R^2 为 0.931–0.993，RMSE 为 0.032–0.060（比例）。2022 相对较弱。

解释边界：田间 LAI 转换 CC 属于校准数据；曲线为冻结参数中心的确定性轨迹，不是后验预测均值或区间。

源表：F02_CC_calibration_pairs.csv, Calibration_daily_growth.csv, Calibration_metrics.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F02_CC_trajectories_EN.png

2. F05_Biomass_pairs_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Figure 5. Paired observed and simulated aboveground dry biomass in (a) 2019, (b) 2020, (c) 2021 and (d) 2022. The dashed line denotes 1:1 agreement. R^2cal = 1 − SSE/SST; RMSE is in t ha^{-1}. The 37 pairs are calibration data, including digitized records with source-specific assumed errors; the plot is not an independent validation.

解析：四年共 37 对；R^2cal 为 0.969–0.997，RMSE 为 0.556–1.252 t ha^{-1}。

解释边界：含数字化生物量及假定误差尺度；所有配对为样本内校准，未构成独立验证。

源表：F05_Biomass_calibration_pairs.csv, Calibration_metrics.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F05_Biomass_pairs_EN.png

3. F07_parameter_intervals_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Figure 7. Marginal distributions of retained stage-specific MCMC samples for eight parameters during 2019–2022. Colored points and bars show medians and the 2.5th–97.5th sample percentiles; gray ranges indicate year-specific prior supports. Stage 2 is conditional on the fixed stage 1 handoff. These intervals neither certify convergence nor demonstrate absence of boundary effects. Boundary-band masses are reported separately.

解析：2019/2020 CCx 在上界 1%支持宽度内的样本质量约 22.14%/23.06%；2020/2021 早期 WP 下界附近约 11.43%/23.65%。

解释边界：2.5%–97.5%为相关保留样本的经验分位区间；阶段二条件于阶段一交接点，不是经认证的八维联合后验。

源表：F07_parameter_intervals_and_boundary_mass.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F07_parameter_intervals_EN.png

4. S02_parameter_boundary_mass_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Supplementary Figure S2. Percentage of retained draws within the lower or upper 1% of each year-specific prior-support width. The band width is a descriptive diagnostic choice, not a significance threshold. No out-of-support draws is distinct from no concentration near a boundary; substantial near-boundary mass is retained and reported.

解析：上/下界邻近带定义为各年支持宽度的 1%；高质量集中见 CCx 及早期 WP。

解释边界：1%带宽是描述选择，不是显著性阈值；不把无越界等同无边界影响。

源表：F07_parameter_intervals_and_boundary_mass.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/
S02_parameter_boundary_mass_EN.png

5. S01_MCMC_stage1_diagnostics_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Supplementary Figure S1. Existing rank-normalized R-hat and bulk ESS summaries for stage 1, recomputed in the previous source audit. The original strict diagnostic

criteria are not met. Retained draw counts are not effective sample sizes. No long-chain extension is performed as part of this figure revision; stage 2 single-ensemble limitations are listed in the sampling table.

解析：既有阶段一未通过原严格 R-hat/ESS 条件；相关保留样本虽多，有效信息量仍受自相关和 ensemble 差异影响。

解释边界：热图只反映已有诊断记录，不是本轮新长链实验。

源表：MCMC_existing_diagnostics.csv, MCMC_existing_sampling_groups.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/S01_MCMC_stage1_diagnostics_EN.png

6. F09_annual_water_budget_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Figure 9. Annual water-budget components for the calibrated DI baseline during 2019–2022. Inflow is irrigation plus precipitation; capillary and groundwater inputs are zero in these runs. Outflow comprises evapotranspiration, runoff, deep percolation and tile drainage. Storage denotes the absolute end-of-season soil-profile plus surface water storage. Water-balance closure is assessed as inflow − outflow = final storage − initial storage, not as inflow − outflow = final storage. Lines connect annual summaries for comparison and do not represent a statistical trend test.

解析：2021 基准季末土壤水约 705.7 mm，比初始增加约 170.0 mm；其他年度初末变化不同。四年水量残差在数值精度范围内。

解释边界：年际连线不是趋势检验。土壤剖面总储水不等于作物根系可利用水。

源表：F09_annual_water_budget.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F09_annual_water_budget_EN.png

7. S03_annual_water_storage_change_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Supplementary Figure S3. Annual inflow, outflow and signed storage change (ΔS = final − initial storage). The three components satisfy inflow − outflow = ΔS within numerical

precision. Positive ΔS indicates net storage gain. This is the change-in-storage companion to the absolute-storage version of Figure 9.

解析：2021 基准季末土壤水约 705.7 mm，比初始增加约 170.0 mm；其他年度初末变化不同。四年水量残差在数值精度范围内。

解释边界：年际连线不是趋势检验。土壤剖面总储水不等于作物根系可利用水。

源表：F09_annual_water_budget.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/S03_annual_water_storage_change_EN.png

8. R01_Sobol_four_outputs_2x2_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Figure R1. Total-effect Sobol indices for (a) grain yield at 13% moisture, (b) aboveground dry biomass, (c) seasonal evapotranspiration and (d) maximum canopy cover. Each cell is the mean of three N=512 scrambles within a year, using the same eight independent uniform parameter ranges. One color scale is used throughout. Rank agreement and variation in index magnitude are assessed separately; indices are not normalized to sum to one. Transpiration and soil evaporation are retained in Supplementary Figure S5.

解析：产量以晚期 WP 和 HI0 为主；生物量以晚期/早期 WP 和 Kcb 为主；ET 以 Kcb 为主；最大 CC 以 CCx 为主。四主输出年度均值排序的 Kendall W 均为 1。

解释边界：只有四个年度模拟情景；2022 产量早期 WP/Kcb 在种子层面近似并列，年度均值相同排序不消除该不确定性。

源表：Sobol_all_seed_indices.csv, Sobol_annual_means_and_ranks.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R01_Sobol_four_outputs_2x2_EN.png

9. S05_Sobol_Tr_Es_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Supplementary Figure S5. Annual mean total-effect indices for transpiration and soil evaporation. All six original model outputs are retained in the source tables; these two outputs supplement the four primary-output panels in Figure R1.

解析：Tr 平均 ST 以 Kcb 约 0.578 为首；Es 以 CCx 约 0.718 为首。两者对同一参数的贡献大小不同。

解释边界：水分分量的 ST 是不同输出方差的归一化比，不能相加成 ET 的 ST。

源表：Sobol_all_seed_indices.csv, Sobol_annual_means_and_ranks.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/S05_Sobol_Tr_Es_EN.png

10. R02_Sobol_Yield_all_years_mean_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Figure R2. Four-year yield sensitivity shown together as (a) first-order and (b) total-effect indices. Colored symbols are annual means across three independent scrambles; black diamonds are equal-weight means across the four annual estimates. Gray segments show the annual minimum–maximum range, not confidence intervals. A four-year mean summarizes these simulation contexts and is not a pooled climate-mixture Sobol index or proof of year-invariant sensitivity. Within-year numerical intervals are retained in Supplementary Figure S4.

解析：晚期 WP、HI0、早期 WP、Kcb 的四年 ST 均值约 0.396、0.206、0.164、0.145；晚期 WP 年度范围约 0.353–0.426。

解释边界：灰线是年际最小—最大范围而非置信区间；先年内平均 3 个 scramble，再四年各 1/4 加权，不等于混合气候下重新计算 Sobol。原图已由用户确认为 R02_Sobol_Yield_CI，按 S1/ST 与数值区间呈现。

源表：Sobol_annual_means_and_ranks.csv, Sobol_four_year_descriptive_means.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/
R02_Sobol_Yield_all_years_mean_EN.png

11. S04_Sobol_yield_numerical_intervals_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Supplementary Figure S4. Total-effect yield indices from three N=512 scrambles per year, with 95% bootstrap half-widths based on 512 resamples. Open diamonds show the within-year scramble mean of S1. The early-WP/Kcb ordering remains numerically uncertain in 2022. This numerical uncertainty is distinct from between-year sensitivity variation. Negative finite-sample S1 values are not set to zero.

解析：N=512、每年 3 个 scramble、512 次 bootstrap；24 项筛查 23 项通过。2022 产量早期 WP 与 Kcb 的 ST 均约 0.139，排名随种子互换。

解释边界：这里的 95%区间不是田间产量预测区间，S1 有限样本负值不强制改零。

源表：Sobol_all_seed_indices.csv, Sobol_numerical_stability.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/
S04_Sobol_yield_numerical_intervals_EN.png

12. R03_full_range_diagnostic_2x2_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Supplemental full-range diagnostic. Solid lines are nondominated best-found actual model evaluations from three NSGA-III searches per year (80 individuals × 60 generations), using frozen yearly parameter centers and four fixed-date irrigation windows. The original 95% observed-yield gate is removed for these searches; the dotted line is shown only as a reference. Gray dashed curves scale the original event amounts proportionally and are not optimized fronts. Open circles show zero irrigation; stars show original DI model baselines. This conditional model reference is separate from field validation and from the gated main optimization.

解析：移除产量门槛的补充搜索共 57,600 次；四年零灌溉产量约 9.071、6.184、6.140、5.850 t ha^{-1}，均明显低于 DI。

解释边界：仅补充诊断取消 95%门槛；仍保留灌水上限，未新增储水约束。没有零灌溉田间验证，也未证明全局最优。

源表：Full_range_best_found_front.csv, Fixed_proportion_dose_response.csv, Zero_irrigation_reference.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/
R03_full_range_diagnostic_2x2_EN.png

13. N01_Optimized_yield_irrigation_EN

阶段：11_English_figures_tables_and_interpretations

English caption：New figure. Best-found full-range yield–irrigation trade-offs from supplemental actual-model NSGA-III searches. Each curve is an upper nondominated envelope, not the response of a single fixed irrigation pattern or a proven global optimum. Zero-irrigation and original DI baselines are shown explicitly. Initial soil water, rainfall, climate and parameter centers remain year specific; terminal storage and per-event delivery limits are not additional constraints in these searches.

解析：移除产量门槛的补充搜索共 57,600 次；四年零灌溉产量约 9.071、6.184、6.140、5.850 t ha^{-1}，均明显低于 DI。

解释边界：仅补充诊断取消 95%门槛；仍保留灌水上限，未新增储水约束。没有零灌溉田间验证，也未证明全局最优。

源表：Full_range_best_found_front.csv, Zero_irrigation_reference.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/N01_Optimized_yield_irrigation_EN.png

14. R04_right_yield_comparison_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Observed yields are the values reported in the source study; DI baseline and selected-candidate yields are simulations at frozen annual parameter centers. Candidates minimize irrigation within the NSGA-III union subject to the DI irrigation cap and Y13 ⩾ 0.95 Yobs. Dotted segments mark this deterministic yield constraint, not confidence intervals. Symbols are offset horizontally within each year for readability. Candidate colors consistently denote 2019, 2020, 2021 and 2022. Lower irrigation does not imply an equal reduction in ET; terminal soil storage is reported separately.

解析：四年候选灌水约 218.36、312.42、267.30、260.78 mm，较 DI 减少 50.96%、44.19%、57.80%、35.58%。产量接近 95%观测门槛。

解释边界：95%虚线不是置信区间。减少灌水不能直接写为等幅减少 ET 或无条件节水不减产。

源表：R04_yield_comparison.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R04_right_yield_comparison_EN.png

15. R04_irrigation_yield_comparison_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Panel (a): percentages are reductions in applied irrigation relative to the original annual DI schedule, not consumptive-water savings or a demonstrated sustainable water-saving rate. Observed yields are the values reported in the source study; DI baseline and selected-candidate yields are simulations at frozen annual parameter centers. Candidates minimize irrigation within the NSGA-III union subject to the DI irrigation cap and Y13 ⩾ 0.95 Yobs. Dotted segments mark this deterministic yield constraint, not confidence intervals.

Symbols are offset horizontally within each year for readability. Candidate colors consistently denote 2019, 2020, 2021 and 2022. Lower irrigation does not imply an equal reduction in ET; terminal soil storage is reported separately.

解析：四年候选灌水约 218.36、312.42、267.30、260.78 mm，较 DI 减少 50.96%、44.19%、57.80%、35.58%。产量接近 95%观测门槛。

解释边界：95%虚线不是置信区间。减少灌水不能直接写为等幅减少 ET 或无条件节水不减产。

源表：R04_yield_comparison.csv, R03_irrigation_ET_storage_tradeoff.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/
R04_irrigation_yield_comparison_EN.png

16. R03_Gated_Pareto_fronts_EN

阶段：11_English_figures_tables_and_interpretations

English caption：English reconstruction of the supplied R03 from its canonical 3,986 final feasible Pareto records. Colored points show NSGA-III results; solid and dashed curves are nondominated unions across five seeds for NSGA-III and NSGA-II, respectively (100 individuals × 100 generations per seed). Each panel uses its own zoomed axes and includes only yield-gated feasible points, not the full dose response. Zero-irrigation yields fall below the gate in every year and therefore are absent. Stars are model baselines, not observations. Open circles are minimum-irrigation selections from the gated NSGA-III union, not geometric knee points. Use the separate full-range figure to interpret the omitted low-irrigation region.

解析：3,986 条最终可行记录；NSGA-II/III 前沿基本重合。2021 可行产量约 15.067–15.251 t ha^{-1}，独立纵轴截取使曲线显得陡。

解释边界：零灌溉被产量门槛排除；圈点不是几何膝点。曲线是搜索得到的非支配包络，未证明全局最优。

源表：Original_main_NSGA_all_seed_Pareto_points.csv, R04_yield_comparison.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R03_Gated_Pareto_fronts_EN.png

17. S07_Irrigation_reduction_decomposition_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Irrigation reduction equals lower ET plus the difference in final soil storage and lower runoff/drainage, because rainfall, initial soil water, other inflows and changes in surface

storage match within each annual baseline–candidate pair. The storage term is the baseline final storage minus candidate final storage. It includes avoided baseline recharge and/or additional depletion and is not wholly attributable to soil-water mining. In 2021, baseline storage increases by about 170.0 mm whereas candidate storage decreases by about 186.7 mm; their 356.7 mm difference accounts for 97.4% of the irrigation reduction. Bars do not demonstrate long-term sustainability; terminal storage was not constrained in the original optimization.

解析：2021 少灌 366.2 mm，其中 ET 仅少 8.6 mm，季末储水差 356.7 mm，解释 97.4%的减灌水量；四年储水项占比约 83%–97%。

解释边界：不能把整个储水差都叫土壤水透支；无季末储水限制的单季方案不能保证跨季可持续。

源表：R03_irrigation_ET_storage_tradeoff.csv, Original_main_Water_ledger_baseline_and_candidate.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/
S07_Irrigation_reduction_decomposition_EN.png

18. Figure1_three_column_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Three-column workflow linking original inputs, grouped conditional MCMC calibration, independent-range Sobol analysis, two-objective irrigation optimization and candidate checks. MCMC samples remain frozen and their diagnostic limitations are retained.

解析：四年原模型已接入；MCMC 冻结；正式 Sobol 为 N=512×3 种子；双算法优化各 100 个体×100 代×5 种子。

解释边界：流程完成不表示采样诊断全部通过，也不表示参数不确定性已在寻优中传播。

源表：

图像：stages/11_English_figures_tables_and_interpretations/figures/Figure1_three_column_EN.png

19. Figure1_tree_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Alternative tree layout of the same four-year workflow. This is an alternative to the three-column version and should not be inserted twice into the main manuscript.

解析：四年原模型已接入；MCMC 冻结；正式 Sobol 为 N=512×3 种子；双算法优化各 100 个体×100 代×5 种子。

解释边界：流程完成不表示采样诊断全部通过，也不表示参数不确定性已在寻优中传播。

源表：

图像：stages/11_English_figures_tables_and_interpretations/figures/Figure1_tree_EN.png

20. Figure10_environment_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Daily reference evapotranspiration and precipitation over the actual model calendar in 2019–2022. Lines and bars retain the exact source dates and values, with consistent axis limits across years. Reference evapotranspiration is distinct from simulated actual crop evapotranspiration.

解析：模型日历合计 599 天；四年降雨分别约 177.9、113.5、143.3、151.4 mm。逐日 ETo 与降雨保持源值，不插值或按年缩放。

解释边界：仅四个年度情景；气象差异与参数、初始水差异并存，不能由本图识别单一气候因果效应。

源表：Figure10_daily_environment.csv, Figure10_window_summary.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/Figure10_environment_EN.png

21. F03_Biomass_trajectories_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Daily aboveground dry biomass at frozen annual parameter centers and the 37 paired calibration observations. These trajectories are deterministic simulations, not posterior predictive means. Errors are in-sample calibration metrics.

解析：四年共 37 对；R^2cal 为 0.969–0.997，RMSE 为 0.556–1.252 t ha^{-1}。

解释边界：含数字化生物量及假定误差尺度；所有配对为样本内校准，未构成独立验证。

源表：Calibration_daily_growth.csv, F05_Biomass_calibration_pairs.csv, Calibration_metrics.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F03_Biomass_trajectories_EN.png

22. F04_CC_pairs_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Observed and simulated canopy cover at the same 48 calibration dates. Dashed lines indicate 1:1 agreement. R^2cal = 1 − SSE/SST is computed against observations without refitting a regression. This is an alternative display to F02, not independent validation.

解析：共 48 对；年度校准 R^2 为 0.931–0.993，RMSE 为 0.032–0.060（比例）。2022 相对较弱。

解释边界：田间 LAI 转换 CC 属于校准数据；曲线为冻结参数中心的确定性轨迹，不是后验预测均值或区间。

源表：F02_CC_calibration_pairs.csv, Calibration_metrics.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F04_CC_pairs_EN.png

23. F06_calibration_summary_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Canopy-cover and biomass RMSE are calculated from paired calibration dates. Grain-yield bias compares the annual model baseline with published observations. The ±5% shaded band is a descriptive project tolerance, not a confidence interval or independent validation.

解析：2022 冠层 RMSE 最高，但生物量 RMSE 最低；不同输出的拟合优劣不必一致。年度籽粒产量偏差约−4.18%至+0.92%。

解释边界：±5%灰带是项目容许幅度，不是统计置信区间；年产量每年只有一个观测终值。

源表：F06_calibration_summary_EN.csv, Calibration_metrics.csv, observed_yield_gate_headroom.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F06_calibration_summary_EN.png

24. F08_phase2_compensation_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Late WP and HI0 are taken from the same retained stage-2 sample rows. Correlations use every retained row; points are evenly thinned for display. Association indicates conditional parameter compensation and is not a causal relationship or proof of convergence.

解析：HI0 与晚期 WP 的 Pearson r 依次为-0.871、-0.595、-0.463、-0.645，均呈负相关。

解释边界：r 使用同一行的全部保留样本计算，散点只为显示抽稀；不推断因果，不用该图认证收敛。

源表：F08_phase2_correlation.csv, F08_displayed_same_row_samples.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F08_phase2_compensation_EN.png

25. F14_phase2_trace_WP_early_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Only the retained segment of the original stage-2 ensemble is shown. The line is the median across 32 walkers at each step; shading is the 10th–90th walker-percentile range, not a posterior confidence band or an independent-chain convergence certificate.

解析：四年均完整保留；线为每步 32 个 walker 的中位数，浅带为 10%–90%分位。图从原步号展示，仅保留段有值。

解释边界：阴影不是参数均值置信区间，也不是独立链间一致性证据；本轮没有延长原链。

源表：F14_F16_phase2_trace_summaries.csv, sampling_group_summary.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F14_phase2_trace_WP_early_EN.png

26. F15_phase2_trace_WP_late_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Only the retained segment of the original stage-2 ensemble is shown. The line is the median across 32 walkers at each step; shading is the 10th–90th walker-percentile range, not a posterior confidence band or an independent-chain convergence certificate.

解析：四年均完整保留；线为每步 32 个 walker 的中位数，浅带为 10%–90%分位。图从原步号展示，仅保留段有值。

解释边界：阴影不是参数均值置信区间，也不是独立链间一致性证据；本轮没有延长原链。

源表：F14_F16_phase2_trace_summaries.csv, sampling_group_summary.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F15_phase2_trace_WP_late_EN.png

27. F16_phase2_trace_HI0_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Only the retained segment of the original stage-2 ensemble is shown. The line is the median across 32 walkers at each step; shading is the 10th–90th walker-percentile range, not a posterior confidence band or an independent-chain convergence certificate.

解析：四年均完整保留；线为每步 32 个 walker 的中位数，浅带为 10%–90%分位。图从原步号展示，仅保留段有值。

解释边界：阴影不是参数均值置信区间，也不是独立链间一致性证据；本轮没有延长原链。

源表：F14_F16_phase2_trace_summaries.csv, sampling_group_summary.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F16_phase2_trace_HI0_EN.png

28. F17_parameter_centers_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Arithmetic means of retained stage-specific parameter samples used as deterministic annual centers. Stage 2 is conditional on the stage-1 handoff; these eight values do not define a certified joint posterior mean. Connecting lines aid annual comparison without establishing temporal trends.

解析：晚期 WP 中心依次为 30.58、25.85、21.94、31.50 g m^{-2}；各年中心与原冻结文件一致。

解释边界：拼接阶段均值不构成完整联合后验的数学均值；年度虚线只是视觉连接。

源表：central_vectors_with_provenance.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F17_parameter_centers_EN.png

29. F11_CC_evidence_roles_EN

阶段：11_English_figures_tables_and_interpretations

English caption：The field LAI-derived DI canopy-cover series enters calibration. Remote-sensing CCquad is converted from percent to fraction and shown only as auxiliary evidence. The 2019–2021 remote-sensing records refer to BI and the 2022 treatment metadata are incomplete; none is relabeled as independent DI validation.

解析：田间 DI 序列用于校准；遥感未入似然，2019–2021 为 BI 标签，2022 处理标签不完整。

解释边界：遥感 CCquad 由百分数转比例；横轴为模型起点后的日数，不能混用未经核实的物候日期。

源表：F11_evidence_series_EN.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F11_CC_evidence_roles_EN.png

30. F12_sampling_record_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Each original ensemble used 32 walkers and 2,000 raw steps. Stage 1 has four ensembles per year and stage 2 has one. The dashed 50-IAT line is only a length reference, not proof of convergence; correlated retained-row counts must not be reported as effective sample sizes.

解析：原 ensemble 均为 32 walkers、2,000 步；阶段一每年 4 个 ensemble，阶段二每年 1 个；阶段一长度不足 50τ 参考。

解释边界：walker 不等于独立链；保留行数不等于 ESS。阶段二不能仅因超过 50τ 就宣称完成严格收敛诊断。

源表：sampling_group_summary.csv, MCMC_existing_diagnostics.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F12_sampling_record_EN.png

31. F09_water_ledger_EN

阶段：11_English_figures_tables_and_interpretations

English caption：English version of the original two-column water ledger, retained as a supplementary alternative to the annual three-line F09. Inflow plus depletion equals outflow plus storage gain. Net storage change is final minus initial storage, not the absolute final storage.

解析：2021 基准季末土壤水约 705.7 mm，比初始增加约 170.0 mm；其他年度初末变化不同。四年水量残差在数值精度范围内。

解释边界：年际连线不是趋势检验。土壤剖面总储水不等于作物根系可利用水。

源表：F09_annual_water_budget.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F09_water_ledger_EN.png

32. F13_yield_gate_EN

阶段：11_English_figures_tables_and_interpretations

English caption：English reconstruction of the original F13 yield-gate figure. The gate is 95% of published observed grain yield. Headroom is the allowable yield decrease relative to the model baseline, not an irrigation-saving rate. This does not assign meanings to the unresolved labels Y_catmul and Y_tot.

解析：2021 模型基准约 15.197，95%观测门槛 15.067 t ha^{-1}，相对模型基准可下降余量仅约 0.86%。

解释边界：这是原 F13 门槛图英文化，尚未把 Y_catmul 或 Y_tot 映射为任何未经确认的列；N02 另给明确变量名的回归图。

源表：observed_yield_gate_headroom.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/F13_yield_gate_EN.png

33. R01_Sobol_ST_Y_13pct_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Annual mean total-effect sensitivity for grain yield at 13% moisture, averaged over three independent N=512 scrambles. All years use the same eight independent uniform parameter ranges and the same 0–1 color scale. Values are not normalized to sum to one. This individual panel is an alternative to the combined R01/S05 displays.

解析：四年等权 ST 描述均值前三位：WP_late 0.396；HI0 0.206；WP_early 0.164。

解释边界：这是公共独立均匀范围的 Sobol，不是相关 MCMC 后验敏感性；ST 不要求加和为 1。本图是合并图的单输出备选。

源表：Sobol_all_seed_indices.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R01_Sobol_ST_Y_13pct_EN.png

34. R01_Sobol_ST_B_t_ha_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Annual mean total-effect sensitivity for aboveground dry biomass, averaged over three independent N=512 scrambles. All years use the same eight independent uniform parameter ranges and the same 0–1 color scale. Values are not normalized to sum to one. This individual panel is an alternative to the combined R01/S05 displays.

解析：四年等权 ST 描述均值前三位：WP_late 0.503；WP_early 0.207；Kcb 0.180。

解释边界：这是公共独立均匀范围的 Sobol，不是相关 MCMC 后验敏感性；ST 不要求加和为 1。本图是合并图的单输出备选。

源表：Sobol_all_seed_indices.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R01_Sobol_ST_B_t_ha_EN.png

35. R01_Sobol_ST_Tr_mm_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Annual mean total-effect sensitivity for seasonal transpiration, averaged over three independent N=512 scrambles. All years use the same eight independent uniform parameter ranges and the same 0–1 color scale. Values are not normalized to sum to one. This individual panel is an alternative to the combined R01/S05 displays.

解析：四年等权 ST 描述均值前三位：Kcb 0.578；CCx 0.227；CGC_CD 0.180。

解释边界：这是公共独立均匀范围的 Sobol，不是相关 MCMC 后验敏感性；ST 不要求加和为 1。本图是合并图的单输出备选。

源表：Sobol_all_seed_indices.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R01_Sobol_ST_Tr_mm_EN.png

36. R01_Sobol_ST_Es_mm_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Annual mean total-effect sensitivity for seasonal soil evaporation, averaged over three independent N=512 scrambles. All years use the same eight independent uniform parameter ranges and the same 0–1 color scale. Values are not normalized to sum to one. This individual panel is an alternative to the combined R01/S05 displays.

解析：四年等权 ST 描述均值前三位：CCx 0.718；CGC_CD 0.267；SenescenceCD 0.022。

解释边界：这是公共独立均匀范围的 Sobol，不是相关 MCMC 后验敏感性；ST 不要求加和为 1。本图是合并图的单输出备选。

源表：Sobol_all_seed_indices.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R01_Sobol_ST_Es_mm_EN.png

37. R01_Sobol_ST_ET_mm_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Annual mean total-effect sensitivity for seasonal evapotranspiration, averaged over three independent N=512 scrambles. All years use the same eight independent uniform parameter ranges and the same 0–1 color scale. Values are not normalized to sum to one. This individual panel is an alternative to the combined R01/S05 displays.

解析：四年等权 ST 描述均值前三位：Kcb 0.837；CGC_CD 0.102；CCx 0.059。

解释边界：这是公共独立均匀范围的 Sobol，不是相关 MCMC 后验敏感性；ST 不要求加和为 1。本图是合并图的单输出备选。

源表：Sobol_all_seed_indices.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R01_Sobol_ST_ET_mm_EN.png

38. R01_Sobol_ST_MaxCC_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Annual mean total-effect sensitivity for maximum canopy cover, averaged over three independent N=512 scrambles. All years use the same eight independent uniform parameter ranges and the same 0–1 color scale. Values are not normalized to sum to one. This individual panel is an alternative to the combined R01/S05 displays.

解析：四年等权 ST 描述均值前三位：CCx 0.911；CGC_CD 0.097；Kcb 0.001。

解释边界：这是公共独立均匀范围的 Sobol，不是相关 MCMC 后验敏感性；ST 不要求加和为 1。本图是合并图的单输出备选。

源表：Sobol_all_seed_indices.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R01_Sobol_ST_MaxCC_EN.png

39. R05_Actual_irrigation_schedules_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Actual event amounts are compared at original annual irrigation dates. Decisions redistribute totals across four ordered windows while preserving baseline proportions within each window. No irrigation dates are optimized. Planned amounts canceled after crop termination are retained separately in the source table.

解析：原事件数为 8、8、7、5；2020 主候选单次最大量约 179.9 mm。日期固定，调整的是窗口总水量及相应事件水量。

解释边界：原始逐次记录才是实际用水，提前结束后取消的计划水量另列；需核实设施和入渗能力才能实施。

源表：NSGA_event_schedules.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/
R05_Actual_irrigation_schedules_EN.png

40. R06_Seeded_convergence_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Each algorithm used 100 individuals × 100 generations × five seeds per year. Curves show seed medians and shading shows minimum–maximum seed ranges, not confidence intervals. Hypervolume uses annual observed yield and DI irrigation for normalization and reference point [0, 1.01]. Compare algorithms within a year; convergence histories do not prove a global optimum.

解析：40 次搜索共 400,000 次评估、4,000 代记录；逐年并集 HV 相对差异绝对值小于 0.07%，没有 NSGA-III 明显优越性的证据。

解释边界：阴影是 5 种子最小—最大范围，不是置信区间。不同年份 HV 归一化背景不同，不横向排名算法性能。

源表：NSGA_convergence_history.csv, NSGA_algorithm_comparison.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R06_Seeded_convergence_EN.png

41. R07_Conditional_candidate_checks_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Each group uses 128 whole-row draws from one retained stage-specific sample pool while fixing the other group at its center. Left intervals describe Monte Carlo sampling uncertainty conditional on that empirical pool, not field failure probability or original MCMC uncertainty. These are not joint eight-parameter posterior draws; 128 perturbations are not 128 effective posterior samples. Right dotted marks are deterministic yield gates.

解析：四个主候选的组内门槛达标比例约 0.281–0.555，Q05 均低于相应门槛；仅中心点达标不代表扰动后高达标率。

解释边界：每次只扰动一个阶段并固定另一个；Wilson 区间仅描述给定经验池的重采样误差，不是田间失败概率，未含原链和模型结构误差。

源表：NSGA_conditional_validation.csv, NSGA_main_candidates.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/
R07_Conditional_candidate_checks_EN.png

42. R09_Terminal_storage_screening_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Dark bars are minimum-irrigation points among already evaluated feasible solutions satisfying final storage at least equal to the original DI baseline; selected points were replayed. This is post hoc filtering, not a new storage-constrained optimization or proof of infeasibility/optimality. The baseline final storage is a sensitivity scenario, not an established agronomic sustainability target.

解析：事后筛选要求季末储水不低于 DI 时，既有点中四年减灌比例约 8.75%、0.56%、0.23%、7.01%，明显小于原主候选。

解释边界：原 DI 季末值不自动等于合理可持续标准；未找到既有点不等于证明不可行。

源表：Terminal_storage_screening.csv, NSGA_main_candidates.csv, Water_ledger_baseline_and_candidate.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/R09_Terminal_storage_screening_EN.png

43. N02_Biomass_regression_EN

阶段：11_English_figures_tables_and_interpretations

English caption：Additional regression using all 37 biomass calibration pairs from 2019–2022. X is simulated and Y is observed aboveground dry biomass, explicitly named rather than mapped to unresolved Y_catmul/Y_tot labels. OLS R^2 is calculated after fitting an intercept and differs from the 1:1 calibration R^2 used in F05. Pearson r describes association. Pooling repeated observations across years can inflate apparent agreement; no independent-observation p-value or validation claim is made.

解析：37 对数据的 OLS R^2=0.9892，Pearson r=0.9946；横轴模拟生物量，纵轴观测生物量。

解释边界：含年内重复日期和跨年合并效应，未给独立样本显著性推断；OLS R^2 与 F05 的 1:1 校准 R^2 不同。未擅自解释 Y_catmul/Y_tot。

源表：F05_Biomass_calibration_pairs.csv, N02_Biomass_regression_summary.csv

图像：stages/11_English_figures_tables_and_interpretations/figures/N02_Biomass_regression_EN.png

44. F13_Pooled_biomass_regression_EN

阶段：11_English_figures_tables_and_interpretations/Final_revision

English caption：Pooled four-year biomass regression, using all 37 calibration pairs. Observed biomass is on the horizontal axis and simulated biomass on the vertical axis. The solid line is an ordinary least-squares fit; the dashed line is 1:1. OLS R^2 and Pearson r describe association, not independent validation; repeated dates occur within years. Digitized biomass retains its source uncertainty.

解析：四年 37 个生物量配对；横轴观测、纵轴模拟，OLS 与 1:1 关系分开。R^2 和 Pearson r 反映配对关联，不是独立验证。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：F05_Biomass_calibration_pairs.csv, Pooled_biomass_OLS_summary.csv

图像：stages/11_English_figures_tables_and_interpretations/Final_revision/figures/F13_Pooled_biomass_regression_EN.png

45. R02_Sobol_Yield_CI_Four_years_and_mean_EN

阶段：11_English_figures_tables_and_interpretations/Final_revision

English caption：Yield sensitivity across 2019–2022. Colored symbols are means over three independently scrambled N=512 Saltelli designs. Error bars are approximate 95% numerical bootstrap intervals for that three-scramble mean, obtained by independently resampling complete A/AB/B base blocks within each scramble (512 resamples; normal half-width 1.96 SD). Diamonds are equal-weight four-year means; gray segments show the across-year minimum–maximum, not a confidence interval. These are independent-uniform-input Sobol estimates, not correlated-posterior indices or field predictive intervals.

解析：四年 S1/ST 及数值 bootstrap 区间合并，保留四年等权均值和年际范围；完整响应块重采样，无新增模型调用。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：Sobol_all_seed_indices.csv, R02_annual_scramble_mean_bootstrap.csv, R02_complete_block_bootstrap_replicates.npz

图像：stages/11_English_figures_tables_and_interpretations/Final_revision/figures/R02_Sobol_Yield_CI_Four_years_and_mean_EN.png

46. R03_Pareto_four_years_annotated_EN

阶段：11_English_figures_tables_and_interpretations/Final_revision

English caption：Four-year full-range, best-found NSGA-III Pareto envelopes. Blue dashed curves join nondominated archived solutions; red dots highlight exact evaluated controls. Labels give rounded allocations (W1, W2, W3, W4; mm), in fixed irrigation-date windows, not soil-moisture thresholds. Stars are original DI model baselines. The 95% observed-yield line is a reference only: it was removed as a constraint for this supplemental full-range search (3 seeds × 80 individuals × 60 generations per year; 57,600 evaluations in total). Four-year historical settings are Zmax=2.30 m and fMulch=0, with 75% cover. The original 400,000-evaluation gated experiment is separate. Curves are finite-search envelopes, not proven optima or independently validated field responses.

解析：按示例的蓝虚线、红点和四窗口水量标签整理。采用四年历史参数的全范围补充搜索，不能混称根深 1m 重校准或正式有门槛搜索。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：Full_range_best_found_front.csv, Pareto_display_points.csv

图像：stages/11_English_figures_tables_and_interpretations/Final_revision/figures/R03_Pareto_four_years_annotated_EN.png

47. R03_Pareto_2019_annotated_EN

阶段：11_English_figures_tables_and_interpretations/Final_revision

English caption：Four-year full-range, best-found NSGA-III Pareto envelopes. Blue dashed curves join nondominated archived solutions; red dots highlight exact evaluated controls. Labels give rounded allocations (W1, W2, W3, W4; mm), in fixed irrigation-date windows, not soil-moisture thresholds. Stars are original DI model baselines. The 95% observed-yield line is a reference only: it was removed as a constraint for this supplemental full-range search (3 seeds × 80 individuals × 60 generations per year; 57,600 evaluations in total). Four-year historical settings are Zmax=2.30 m and fMulch=0, with 75% cover. The original 400,000-evaluation gated experiment is separate. Curves are finite-search envelopes, not proven optima or independently validated field responses.

解析：按示例的蓝虚线、红点和四窗口水量标签整理。采用四年历史参数的全范围补充搜索，不能混称根深 1m 重校准或正式有门槛搜索。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：Pareto_display_points.csv

图像：stages/11_English_figures_tables_and_interpretations/Final_revision/figures/R03_Pareto_2019_annotated_EN.png

48. R03_Pareto_2020_annotated_EN

阶段：11_English_figures_tables_and_interpretations/Final_revision

English caption：Four-year full-range, best-found NSGA-III Pareto envelopes. Blue dashed curves join nondominated archived solutions; red dots highlight exact evaluated controls. Labels give rounded allocations (W1, W2, W3, W4; mm), in fixed irrigation-date windows, not soil-moisture thresholds. Stars are original DI model baselines. The 95% observed-yield line is a reference only: it was removed as a constraint for this supplemental full-range search (3 seeds × 80 individuals × 60 generations per year; 57,600 evaluations in total). Four-year historical settings are Zmax=2.30 m and fMulch=0, with 75% cover. The original 400,000-evaluation gated experiment is separate. Curves are finite-search envelopes, not proven optima or independently validated field responses.

解析：按示例的蓝虚线、红点和四窗口水量标签整理。采用四年历史参数的全范围补充搜索，不能混称根深 1m 重校准或正式有门槛搜索。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：Pareto_display_points.csv

图像：stages/11_English_figures_tables_and_interpretations/Final_revision/figures/R03_Pareto_2020_annotated_EN.png

49. R03_Pareto_2021_annotated_EN

阶段：11_English_figures_tables_and_interpretations/Final_revision

English caption：Four-year full-range, best-found NSGA-III Pareto envelopes. Blue dashed curves join nondominated archived solutions; red dots highlight exact evaluated controls. Labels give rounded allocations (W1, W2, W3, W4; mm), in fixed irrigation-date windows, not soil-moisture thresholds. Stars are original DI model baselines. The 95% observed-yield line is a

reference only: it was removed as a constraint for this supplemental full-range search (3 seeds × 80 individuals × 60 generations per year; 57,600 evaluations in total). Four-year historical settings are Zmax=2.30 m and fMulch=0, with 75% cover. The original 400,000-evaluation gated experiment is separate. Curves are finite-search envelopes, not proven optima or independently validated field responses.

解析：按示例的蓝虚线、红点和四窗口水量标签整理。采用四年历史参数的全范围补充搜索，不能混称根深 1m 重校准或正式有门槛搜索。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：Pareto_display_points.csv

图像：stages/11_English_figures_tables_and_interpretations/Final_revision/figures/R03_Pareto_2021_annotated_EN.png

50. R03_Pareto_2022_annotated_EN

阶段：11_English_figures_tables_and_interpretations/Final_revision

English caption：Four-year full-range, best-found NSGA-III Pareto envelopes. Blue dashed curves join nondominated archived solutions; red dots highlight exact evaluated controls. Labels give rounded allocations (W1, W2, W3, W4; mm), in fixed irrigation-date windows, not soil-moisture thresholds. Stars are original DI model baselines. The 95% observed-yield line is a reference only: it was removed as a constraint for this supplemental full-range search (3 seeds × 80 individuals × 60 generations per year; 57,600 evaluations in total). Four-year historical settings are Zmax=2.30 m and fMulch=0, with 75% cover. The original 400,000-evaluation gated experiment is separate. Curves are finite-search envelopes, not proven optima or independently validated field responses.

解析：按示例的蓝虚线、红点和四窗口水量标签整理。采用四年历史参数的全范围补充搜索，不能混称根深 1m 重校准或正式有门槛搜索。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：Pareto_display_points.csv

图像：stages/11_English_figures_tables_and_interpretations/Final_revision/figures/R03_Pareto_2022_annotated_EN.png

51. R1_01_Four_year_yield_irrigation

阶段：06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial

English caption：Four-year best-found yield–irrigation trade-offs with maximum root depth 1 m. See the associated source tables and stage report for precise definitions.

解析：用户要求的四年合并产量—灌水量图。空心圆为零灌水，星号为 1m 根深下原 DI 日程的模型回放，空心方框为文献 DI 观测产量。

解释边界：折线连接实际评价得到的非支配点；不是固定比例减灌曲线，也不证明全局最优。原 MCMC 参数冻结，未在 1m 根深下重校准。

源表：Root1m_best_found_front.csv, Baseline_and_zero_replays.csv, Thesis_DI_observed_summary.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial/figures/R1_01_Four_year_yield_irrigation.png

52. R1_02_Pareto_fronts_2x2

阶段：06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial

English caption：Four-year NSGA-III Pareto fronts under a 1 m maximum root depth. See the associated source tables and stage report for precise definitions.

解析：用户要求的四年 2×2 帕累托图，灰点为实际搜索记录的显示抽样，彩色前沿取各年全部五种子、所有代的实际评价并集；星号为 1m 模型原 DI 日程，空心圆为零灌水。

解释边界：模型优化只有产量与实际灌水量两个目标。95%产量线仅作参照，未用于截断搜索；显示抽样不影响前沿计算。

源表：Root1m_best_found_front.csv, Actual_evaluation_file_index.csv, Thesis_DI_observed_summary.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial/figures/R1_02_Pareto_fronts_2x2.png

53. R1_03_Root_depth_and_allocation

阶段：06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial

English caption：Root-depth effects separated from irrigation-allocation effects. See the associated source tables and stage report for precise definitions.

解析：两条比例减灌曲线采用完全相同的日期、窗口比例和总量，仅改变根深；与新 1m 前沿的差别反映重新分配灌水的收益。

解释边界：固定比例路径不是优化前沿；此图不把旧 2.3m 不同搜索预算的前沿当成严格配对实验。

源表：Root1m_best_found_front.csv, Paired_fixed_proportion_response.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial/figures/R1_03_Root_depth_and_allocation.png

54. R1_04_Matched_yield_and_observations

阶段：06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial

English caption：Matched-model-yield irrigation reductions and observed-yield comparison. See the associated source tables and stage report for precise definitions.

解析：左图以 1m 根深下原 DI 模型产量为相同目标，右图单列文献实测与模型最高找到产量，避免把同模型产量说成同实测产量。

解释边界：减灌比例仍可能包含土壤耗储；未设置季末储水约束，不能直接等同耗水减少或田间推荐节水率。

源表：Selected_root1m_front_points.csv, Four_year_root1m_summary.csv, Thesis_DI_observed_summary.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial/figures/R1_04_Matched_yield_and_observations.png

55. R1_05_Selected_irrigation_schedules

阶段：06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial

English caption：Actual event schedules at matched 1 m DI model yield. See the associated source tables and stage report for precise definitions.

解析：保留文献原 DI 灌水日期，比较原水量与新候选实际执行量；四个优化窗口内的原事件比例保持不变。

解释边界：原执行上限不是设备极限，本根深单因素试验未新加硬件事件限制；大单次量的现场可实施性仍需核实。

源表：Selected_actual_event_schedules.csv, Thesis_DI_irrigation_events.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial/figures/R1_05_Selected_irrigation_schedules.png

56. R1_06_Seeded_convergence

阶段：06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial

English caption：Five-seed convergence records for the root-depth experiment. See the associated source tables and stage report for precise definitions.

解析：展示五种子各代种群超体积、种子中位数与范围，数据来自完整 100 代记录。

解释边界：阴影不是置信区间；当前种群指标可波动，指标稳定也不证明全局最优。前沿计算另保留所有已评价非支配点。

源表：Convergence_all_seeds.csv, Seed_stability_summary.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Root1m_frozen_4year_trial/figures/R1_06_Seeded_convergence.png

57. P01_CC_fit_2021_root1m

阶段：07_2021_root1m_pilots/Calibration

English caption：2021 canopy-cover fit with maximum root depth fixed at 1 m. See the associated source tables and stage report for precise definitions.

解析：左图将原起点和新校准试跑的逐日 CC 与田间 LAI 换算 CC 配对；右图展示实测与模拟的一致性，并列出 RMSE、拟合 R^2 和 Pearson r。

解释边界：CC 是校准资料而非独立验证；未把短链分布当作正式可信区间。拟合 R^2=1−SSE/SST，与 Pearson r^2 不同。

源表：Phase2_pilot_CC_fitted_pairs.csv, Phase2_pilot_daily_growth.csv, Fit_metrics.csv

图像：stages/07_2021_root1m_pilots/Calibration/figures/P01_CC_fit_2021_root1m.png

58. P02_Yield_and_biomass

阶段：07_2021_root1m_pilots/Calibration

English caption：Yield agreement and conditional biomass fit for the 2021 pilot. See the associated source tables and stage report for precise definitions.

解析：左图比较 1m 根深下原起点与两阶段试跑最终参数的产量，并以文献 15.86t/ha 标出差异；右图给出条件阶段二的生物量拟合。

解释边界：产量进入阶段二似然，属于样本内拟合；生物量误差条是原代码假设 sigma，不是实测标准差。原内部产量靶值 15.862t/ha 与论文显示精度 15.86 同时保留。

源表：Yield_and_water_comparison.csv, Phase2_pilot_Biomass_fitted_pairs.csv

图像：stages/07_2021_root1m_pilots/Calibration/figures/P02_Yield_and_biomass.png

59. P03_Parameter_boundary_screen

阶段：07_2021_root1m_pilots/Calibration

English caption：Parameter-boundary screening of eight calibrated pilot parameters. See the associated source tables and stage report for precise definitions.

解析：八个面板横轴完整显示先验范围；两侧淡橙色为各 1%边界带，橙线为选中的短链样本点；角标给出边界带内样本质量。

解释边界：分布来自未确认收敛的短链，不能凭图宣称无撞墙。需同时查看选点距离、1%/5%边界质量、跨组 Rhat 与 ESS；SenescenceCD 在引擎内按原代码取整。

源表：All_parameter_boundary_and_mixing_checks.csv, runs/*/pilot_retained_draws_last250.csv

图像：stages/07_2021_root1m_pilots/Calibration/figures/P03_Parameter_boundary_screen.png

60. P04_MCMC_trace_and_mixing

阶段：07_2021_root1m_pilots/Calibration

English caption：Short-chain traces and matched-walker mixing diagnostics. See the associated source tables and stage report for precise definitions.

解析：淡线为各 walker，粗线为同组 walker 均值；四种颜色为独立 ensemble，灰区为试跑摘要排除的前 250 步。标题给出跨独立组匹配 walker 诊断的保守 Rhat 与 ESS。

解释边界：同组 walker 相互耦合，未当成独立链。500 步不满足原 2000 步正式要求；即便拟合好也不能自动认定收敛。

源表：runs/canopy_transpiration/pilot_arrays.npz, runs/biomass_yield/pilot_arrays.npz, All_parameter_boundary_and_mixing_checks.csv

图像：stages/07_2021_root1m_pilots/Calibration/figures/P04_MCMC_trace_and_mixing.png

61. P05_Yield_irrigation_0_to_actual

阶段：07_2021_root1m_pilots/Calibration

English caption：2021 yield–irrigation curve from zero to the actual DI amount. See the associated source tables and stage report for precise definitions.

解析：采用本轮新校准参数逐点计算 201 个水量水平；灌水日期和原逐次比例不变，横轴用引擎实际执行量，严格从 0 到 633.47mm，终点与新拟合 DI 回放一致。

解释边界：此图是固定比例剂量响应，不是 NSGA-III 帕累托前沿；零灌水仍有初始土壤水与 143.32mm 季节降雨，所以不强制零产量。低水量尚无独立田间验证。

源表：Yield_irrigation_curve_0_to_observed.csv, Yield_irrigation_curve_actual_event_amounts.csv

图像：stages/07_2021_root1m_pilots/Calibration/figures/P05_Yield_irrigation_0_to_actual.png

62. N21_01_Pareto_front

阶段：07_2021_root1m_pilots/NSGA3

English caption：2021 NSGA-III yield–irrigation Pareto front with soil-storage audit. See the associated source tables and stage report for precise definitions.

解析：以全部 30,000 次实际优化评价的非支配并集构成前沿；灰点为显示抽样，虚线为原 DI 同比例减灌响应。前沿颜色代表季末 0—1m 储水，星号与方框分开表示原 DI 模型和文献实测。

解释边界：二维目标为产量最大与实际灌水最少，颜色只是审计量，不是第三目标。前沿不强行延长到 633.47mm，被支配的原 DI 仍保留参照。固定参数来自未正式收敛的短链；未设置产量门槛或新增储水约束。

源表：NSGA3_best_found_front.csv, All_NSGA3_evaluations.csv.gz, Baseline_and_zero.csv, sources/Proportional_DI_response.csv

图像：stages/07_2021_root1m_pilots/NSGA3/figures/N21_01_Pareto_front.png

63. N21_02_Storage_qualification

阶段：07_2021_root1m_pilots/NSGA3

English caption：Soil-storage qualification and its effect on the evaluated trade-off. See the associated source tables and stage report for precise definitions.

解析：蓝线是未附加储水约束的主前沿；橙线从所有已评价候选中筛出季末 0—1m 储水不低于原 DI 者，再取非支配点。右图单独展示储水，避免把根层耗储当作无代价节水。

解释边界：原 DI 季末储水是同模型比较基准，允许 0.01mm 报告精度容差，不是新增实测阈值。橙线不是另一次带约束 NSGA-III 的优化结果，也不证明已充分搜索该子域；候选集合与主前沿可能不同。

源表：NSGA3_best_found_front.csv, Storage_qualified_evaluated_subset_front.csv, Selected_front_solutions.csv

图像：stages/07_2021_root1m_pilots/NSGA3/figures/N21_02_Storage_qualification.png

64. N21_03_Seed_stability

阶段：07_2021_root1m_pilots/NSGA3

English caption：Three-seed front agreement and NSGA-III search progress. See the associated source tables and stage report for precise definitions.

解析：三个独立随机种子各运行 100 个体、100 代；左图比较各自全部评价的非支配前沿，右图为每代当前种群的归一化超体积。

解释边界：种子差异不是统计置信区间；种群超体积稳定也不证明全局最优。前沿并集使用所有代记录，不只取最后一代。

源表：Seed_stability.csv, Generation_history_all_seeds.csv, runs/seed*/all_evaluated_front.csv

图像：stages/07_2021_root1m_pilots/NSGA3/figures/N21_03_Seed_stability.png

65. W01_Water_stress_and_canopy

阶段：07_2021_root1m_pilots/Water_audit

English caption：Water stress and canopy responses under five irrigation amounts. See the associated source tables and stage report for precise definitions.

解析：前三面板分别显示冠层扩展、气孔和衰老的模型缺水胁迫系数；1 表示不受该项缺水限制，越低胁迫越强。第四面板给出冠层响应，实测 CC 仅对应原 DI 条件。细灰竖线是原灌水日期。

解释边界：读取真实函数调用；气孔面板使用本版本蒸腾实际采用的 sto_lin。扩展只显示模型扩展期且非种子保护期，气孔只显示潜在蒸腾大于零的日子，衰老只显示非保护期且仍有绿冠层的日子。断线不补成无胁迫；系数不等于独立观测或百分比减产。

源表：All_scenarios_daily.csv, Scenario_water_and_stress_summary.csv, I*_stress_function_calls.csv

图像：stages/07_2021_root1m_pilots/Water_audit/figures/W01_Water_stress_and_canopy.png

66. W02_Root_domain_and_deep_storage

阶段：07_2021_root1m_pilots/Water_audit

English caption：Root-domain storage and water retained below the maximum root depth. See the associated source tables and stage report for precise definitions.

解析：上图统一采用 0—1m 固定土层储水，横向比较各灌水情景，虚线为模型田持水量与萎蔫点对应储水；下图单列 1—2.4m 储水的相对初值变化，避免把根区以下存水当作根系可用水。

解释边界：日期是日末状态对应的次日零时，含 4 月 27 日模拟前初值。根深动态增长，0—1m 固定层不等于幼苗期实际可达根域。FC/WP 来自原质地转换；初值来自近似重建。深层没有根系直接取水，底边界位于 2.4m。

源表：All_scenarios_daily.csv, Initial_water_hydraulic_source_audit.csv, Scenario_water_and_stress_summary.csv

图像：stages/07_2021_root1m_pilots/Water_audit/figures/W02_Root_domain_and_deep_storage.png

67. F01_DI_canopy_biomass_fits

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Canopy and biomass fits after conditional recalibration. See the associated source tables and stage report for precise definitions.

解析：三种 fMulch 分别重校准两阶段参数。比较冠层与生物量的残差，同时查看产量偏差；不能仅因产量接近而判定物理结构正确。

解释边界：圆点是原 DI 校准数据，生物量与部分 CC 来自原项目数字化或换算；这是条件拟合，不是独立验证。

源表：fm*_CC_pairs.csv, fm*_Biomass_pairs.csv, fm*_DI_growth.csv, Calibration_fit_metrics.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/F01_DI_canopy_biomass_fits.png

68. F02_DI_yield_Es_Tr_ET

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：DI yield and seasonal water-flux fits. See the associated source tables and stage report for precise definitions.

解析：充分灌溉下产量可能接近，但 Es、Tr、ET 未必同时接近原目标；用共同目标和相同误差权重比较，而非挑选节水幅度符合预期的情景。

解释边界：Es=78mm 是原项目近似重建的校准目标，Tr 由 ET 减 Es 得到，并非新增加的独立实测。不同目标并不相互独立。

源表：Calibrated_DI_fit_summary.csv, Descriptive_fit_residual_scores.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/F02_DI_yield_Es_Tr_ET.png

69. T01_Yield_irrigation_comparison

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Proportional irrigation responses across three mulch structures. See the associated source tables and stage report for precise definitions.

解析：每条线用该地膜情景各自重校准的参数，按原 DI 日程比例缩放灌水；比较 0、300、350、400 及 633.47mm 的产量与储水，不强制曲线在 400mm 转折。

解释边界：这些是单个短链选定参数的剂量响应，既非优化前沿，也没有传播参数后验不确定性；零灌溉允许降雨和初始储水。历史 f0 只作旧结果对照。

源表：All_proportional_responses.csv, Exact_300_350_400_response_checks.csv, Historical_f0_proportional_response.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/T01_Yield_irrigation_comparison.png

70. F03_MCMC_walls_and_mixing

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Parameter bounds and short-chain mixing diagnostics. See the associated source tables and stage report for precise definitions.

解析：检查选定点距先验上下界、边界附近样本占比、Rhat 和 ESS。拟合良好并不代表参数可辨识；WP 或 HI0 贴边时应优先复核先验与模型结构。

解释边界：500 步仅为用户指定的调试试验。保留段分位数不是收敛后验区间，正式原流程最少 2000 步等门槛未通过，不可把 12 万保存状态称为独立有效样本。

源表：All_MCMC_parameter_diagnostics.csv, calibration/*/*/Pilot_arrays.npz

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/F03_MCMC_walls_and_mixing.png

71. T02_fm050_Water_stress_canopy

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Water-stress and canopy responses, fMulch=0.50. See the associated source tables and stage report for precise definitions.

解析：前三面板分别追溯扩展、气孔、衰老的缺水限制，第四面板显示 CC。重点查看 316.74 与 400mm 是否在关键时期维持蒸腾以及冠层。

解释边界：显示实际函数调用；气孔使用本版本实际作用于 Tr 的 sto_lin。按过程适用时间掩膜，断线不填成 1。DI 观测圆点仅对应 633.47mm，不能直接验证缺水情景。

源表：fm050_I*_daily.csv, fm050_I*_stress_function_calls.csv, All_stress_storage_summaries.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/T02_fm050_Water_stress_canopy.png

72. S01_fm050_Soil_water_storage

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Root and deep soil storage, fMulch=0.50. See the associated source tables and stage report for precise definitions.

解析：把根区储水与根区以下储水分开，检查少灌是否靠多消耗土壤水维持产量。季末储水差异是一次性节水与持续管理方案区别的重要依据。

解释边界：水力剖面仍为原 2.4m；根系最大 1m，1m 以下没有直接根系吸水。初始含水率来自原近似重建，未经新增独立校正。

源表：fm050_I*_daily.csv, All_stress_storage_summaries.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/S01_fm050_Soil_water_storage.png

73. T02_fm075_Water_stress_canopy

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Water-stress and canopy responses, fMulch=0.75. See the associated source tables and stage report for precise definitions.

解析：前三面板分别追溯扩展、气孔、衰老的缺水限制，第四面板显示 CC。重点查看 316.74 与 400mm 是否在关键时期维持蒸腾以及冠层。

解释边界：显示实际函数调用；气孔使用本版本实际作用于 Tr 的 sto_lin。按过程适用时间掩膜，断线不填成 1。DI 观测圆点仅对应 633.47mm，不能直接验证缺水情景。

源表：fm075_I*_daily.csv, fm075_I*_stress_function_calls.csv, All_stress_storage_summaries.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/T02_fm075_Water_stress_canopy.png

74. S01_fm075_Soil_water_storage

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Root and deep soil storage, fMulch=0.75. See the associated source tables and stage report for precise definitions.

解析：把根区储水与根区以下储水分开，检查少灌是否靠多消耗土壤水维持产量。季末储水差异是一次性节水与持续管理方案区别的重要依据。

解释边界：水力剖面仍为原 2.4m；根系最大 1m，1m 以下没有直接根系吸水。初始含水率来自原近似重建，未经新增独立校正。

源表：fm075_I*_daily.csv, All_stress_storage_summaries.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/S01_fm075_Soil_water_storage.png

75. T02_fm085_Water_stress_canopy

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Water-stress and canopy responses, fMulch=0.85. See the associated source tables and stage report for precise definitions.

解析：前三面板分别追溯扩展、气孔、衰老的缺水限制，第四面板显示 CC。重点查看 316.74 与 400mm 是否在关键时期维持蒸腾以及冠层。

解释边界：显示实际函数调用；气孔使用本版本实际作用于 Tr 的 sto_lin。按过程适用时间掩膜，断线不填成 1。DI 观测圆点仅对应 633.47mm，不能直接验证缺水情景。

源表：fm085_I*_daily.csv, fm085_I*_stress_function_calls.csv, All_stress_storage_summaries.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/T02_fm085_Water_stress_canopy.png

76. S01_fm085_Soil_water_storage

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Root and deep soil storage, fMulch=0.85. See the associated source tables and stage report for precise definitions.

解析：把根区储水与根区以下储水分开，检查少灌是否靠多消耗土壤水维持产量。季末储水差异是一次性节水与持续管理方案区别的重要依据。

解释边界：水力剖面仍为原 2.4m；根系最大 1m，1m 以下没有直接根系吸水。初始含水率来自原近似重建，未经新增独立校正。

源表：fm085_I*_daily.csv, All_stress_storage_summaries.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/S01_fm085_Soil_water_storage.png

77. T03_all_NSGA3_front

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：NSGA-III best-found yield–irrigation front, all. See the associated source tables and stage report for precise definitions.

解析：实线来自实际评估候选的非支配集合，点虚线是原比例日程。用共同的实测产量和 95%实测产量参照比较，另报告各情景自己的 DI 基线，避免因基线变差而显得更省水。

解释边界：两个目标：产量最大化、实际季灌水最小化；四个固定时间窗分配量。无强制 400mm 下界、无产量门槛、未加入季末储水约束。有限搜索不是已证明的全局最优，也不是田间验证。

源表：All_scenario_NSGA3_fronts.csv, Selected_front_comparisons.csv, All_proportional_responses.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/T03_all_NSGA3_front.png

78. T03_fm050_NSGA3_front

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：NSGA-III best-found yield–irrigation front, fm050. See the associated source tables and stage report for precise definitions.

解析：实线来自实际评估候选的非支配集合，点虚线是原比例日程。用共同的实测产量和 95%实测产量参照比较，另报告各情景自己的 DI 基线，避免因基线变差而显得更省水。

解释边界：两个目标：产量最大化、实际季灌水最小化；四个固定时间窗分配量。无强制 400mm 下界、无产量门槛、未加入季末储水约束。有限搜索不是已证明的全局最优，也不是田间验证。

源表：All_scenario_NSGA3_fronts.csv, Selected_front_comparisons.csv, All_proportional_responses.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/T03_fm050_NSGA3_front.png

79. T03_fm075_NSGA3_front

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：NSGA-III best-found yield–irrigation front, fm075. See the associated source tables and stage report for precise definitions.

解析：实线来自实际评估候选的非支配集合，点虚线是原比例日程。用共同的实测产量和 95%实测产量参照比较，另报告各情景自己的 DI 基线，避免因基线变差而显得更省水。

解释边界：两个目标：产量最大化、实际季灌水最小化；四个固定时间窗分配量。无强制 400mm 下界、无产量门槛、未加入季末储水约束。有限搜索不是已证明的全局最优，也不是田间验证。

源表：All_scenario_NSGA3_fronts.csv, Selected_front_comparisons.csv, All_proportional_responses.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/T03_fm075_NSGA3_front.png

80. T03_fm085_NSGA3_front

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：NSGA-III best-found yield–irrigation front, fm085. See the associated source tables and stage report for precise definitions.

解析：实线来自实际评估候选的非支配集合，点虚线是原比例日程。用共同的实测产量和 95%实测产量参照比较，另报告各情景自己的 DI 基线，避免因基线变差而显得更省水。

解释边界：两个目标：产量最大化、实际季灌水最小化；四个固定时间窗分配量。无强制 400mm 下界、无产量门槛、未加入季末储水约束。有限搜索不是已证明的全局最优，也不是田间验证。

源表：All_scenario_NSGA3_fronts.csv, Selected_front_comparisons.csv, All_proportional_responses.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/T03_fm085_NSGA3_front.png

81. D01_400mm_cross_structure

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Stress, canopy and storage at a shared 400 mm schedule. See the associated source tables and stage report for precise definitions.

解析：固定灌溉总量与日程为 400mm，横向比较地膜结构及其条件重校准的联合作用；观察胁迫、CC 和储水的变化能否解释平台移动。

解释边界：每个情景参数不同，因而不是固定参数的纯 fMulch 导数。原 DI 的 CC 观测不叠加到 400mm 面板，避免误当缺水验证。

源表：fm*_I400_daily.csv, Selected_parameters_all_scenarios.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/D01_400mm_cross_structure.png

82. D02_Storage_qualified_subset

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Effect of post-filtering for comparable terminal root-zone storage. See the associated source tables and stage report for precise definitions.

解析：按各自 DI 季末 0—1m 储水后筛选，查看主前沿少灌的收益有多少依赖额外耗储；原 DI 产量线只用于同一面板比较，三情景仍应另用共同实测门槛。

解释边界：紫虚线只来自主搜索已经评估过的合格候选，并非另一次充分的储水约束搜索；0.01mm 是报告精度容差，不是实测阈值。

源表：All_storage_qualified_subset_fronts.csv, Selected_front_comparisons.csv, Optimization_scenario_summary.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/D02_Storage_qualified_subset.png

83. D03_Applied_water_reduction_partition

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Water-balance partition of reducing irrigation from633.47 to400 mm. See the associated source tables and stage report for precise definitions.

解析：在相同参数、天气和初值下，从 633.47 减到 400mm 少灌 233.47mm，ET 几乎不变；主要差额表现为季末土壤存水减少，尤其 1m 根层以下。这个质量守恒分解比只讨论地膜系数更能解释模型平台。

解释边界：季末根层以下存水不是可直接被根系利用的水；分量来自冻结模型而非实测去向。改变土壤域、水力参数或初值可能改变分配，不能直接把深层存水差叫田间浪费。

源表：Irrigation_reduction_partition_400mm.csv, Exact_300_350_400_response_checks.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/D03_Applied_water_reduction_partition.png

84. D04_Optimized_vs_same_amount_schedule

阶段：08_2021_mulch_conditional_calibration_and_optimization

English caption：Why allocation improves the best-found front. See the associated source tables and stage report for precise definitions.

解析：在相同地膜结构和同一实际灌水总量下，对照优化分配与原比例分配，追溯气孔胁迫、冠层以及季末储水。灰线为原 DI；同时检查四个日历窗口账本和七次事件水量。

解释边界：优化保持原日期，改变四个窗口总量，并非优化灌溉日期。95%目标候选与原 DI 不是同产量的实测田间对照；日历会计窗口也不是重新识别的生育阶段。

源表：Optimized_candidate_stress_summaries.csv, fm*_Opt95_daily.csv, fm*_PropSameOpt95_daily.csv, Calendar_window_water_stress_ledger.csv, Selected_event_schedules.csv

图像：stages/08_2021_mulch_conditional_calibration_and_optimization/figures/D04_Optimized_vs_same_amount_schedule.png

85. D01_Water_accounting_four_years

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Apparent irrigation reductions and soil-storage accounting. See the associated source tables and stage report for precise definitions.

解析：2020/2021 年季末储水差解释灌水减少量的 93.3%/97.4%；原 DI 增储而 C1 耗储。

解释边界：季末储水差不是全部来自初始深层水；该图不证明盆地尺度净节水。

源表：Water_reduction_accounting.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D01_Water_accounting_four_years.png

86. D02_Daily_storage_canopy_stress

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Daily water storage, canopy and stomatal response. See the associated source tables and stage report for precise definitions.

解析：C1 的主要差异出现在季末储水；冠层峰值接近，模型开花窗口内授粉胁迫系数接近 1。

解释边界：灰色窗为模型日历开花窗口，并非独立田间物候观测；K=1 表示该项水分胁迫未触发。

源表：All_daily_traces.csv.gz, Canopy_stress_diagnostics.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D02_Daily_storage_canopy_stress.png

87. D03_Layer_moisture_and_roots

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Soil moisture redistribution and modeled root depth. See the associated source tables and stage report for precise definitions.

解析：深层在原 DI 中有显著补蓄；C1 减轻补蓄，同时降低表层含水。

解释边界：色标为田间持水量与萎蔫点之间的相对可利用水，不是饱和度；色标截在 0 与 1.2；深层土壤性质与初始水仍需现场证据。

源表：All_layer_traces.csv.gz, All_daily_traces.csv.gz

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D03_Layer_moisture_and_roots.png

88. D04_Period_water_budgets

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Water allocation over model calendar periods. See the associated source tables and stage report for precise definitions.

解析：2020 年 C1 几乎全部灌水前置，生殖期依赖储水；2021 年原 DI 后期灌水相当一部分留在土壤。

解释边界：五个诊断日历时期与四个优化决策窗口不同；不得直接解释成五次田间灌水。

源表：Period_water_canopy_stress.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D04_Period_water_budgets.png

89. D05_Event_allocation_and_original_setting

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Actual irrigation events and the original per-event setting. See the associated source tables and stage report for precise definitions.

解析：原接口随候选调大 MaxIrr，2020 年 C1 单次约 180mm 超过原日程执行设置 84.88mm。

解释边界：用户已确认原 MaxIrr 是保障实测日程执行的数值上限，并非设备极限；固定原上限只是受控诊断，不能凭此认定原接口错误。

源表：Actual_event_schedules_DI_C1.csv, Original_configuration.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D05_Event_allocation_and_original_setting.png

90. D06_Root_depth_hypothesis_test

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Root-depth sensitivity of fixed irrigation schedules. See the associated source tables and stage report for precise definitions.

解析：原 DI 拟合良好并不能保证亏缺灌溉外推合理；2020 年 C1 对 2.3m 根深假设非常敏感。

解释边界：缩短根深仅是单因素诊断，未重校准；不能选一个更悲观根深充当真实根深。

源表：Root_depth_hypothesis_test.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D06_Root_depth_hypothesis_test.png

91. D07_Depth_uptake_and_storage_change

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Depth-resolved uptake and net soil-water storage change. See the associated source tables and stage report for precise definitions.

解析：2020 年 C1 在 1m 以下取水 152.8mm，但深层季净储水仍增加 22.2mm；不能把深层取水量全部认定为开采初始水。

解释边界：逐层取水由原引擎蒸腾步骤前后含水差直接记录；未做雨水、灌水、初始水示踪。

源表：Depth_uptake_and_net_storage.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D07_Depth_uptake_and_storage_change.png

92. D08_Constrained_search_comparison

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Diagnostic constrained searches and soil-water carryover. See the associated source tables and stage report for precise definitions.

解析：控制原单次灌水设置后，若不允许消耗初始储水，2020/2021 年最少灌水候选的降幅收敛至 12.4%/27.4%。

解释边界：每年每策略两种子×64 个体×50 代；黑短线表示种子候选，不是置信区间。仍采用 95%观测产量门槛，实际 ET 可响应变化。

源表：Constrained_best_found_candidates.csv, Constrained_seed_candidates.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D08_Constrained_search_comparison.png

93. D09_Equal_ET_diagnostics

阶段：06_Structural_and_root_depth_diagnostics/Mechanism_audit

English caption：Equal actual-ET comparisons with fixed per-event settings. See the associated source tables and stage report for precise definitions.

解析：将季节实际 ET 精确匹配原 DI 后，兼顾季末不低于初始储水的诊断点降幅为 11.1%/26.8%；保持原 DI 季末储水时回到原 DI。

解释边界：这些是经模型求根得到的可行对照点，不是新的最小灌水全局最优；只匹配季总 ET，不强制每日 ET 轨迹相同。

源表：Equal_ET_fixed_management_diagnostics.csv

图像：stages/06_Structural_and_root_depth_diagnostics/Mechanism_audit/figures/D09_Equal_ET_diagnostics.png

94. P01_Daily_precipitation_4years_EN

阶段：09_Rainfall_and_data_provenance_audit

English caption：Blue bars show uploaded GEE inputs; orange bars are approximate reconstructions of thesis Figure 2-2 (PDF p. 22). Pale shading denotes the published DI crop period. Broken rainfall axes in the source for 2021 and 2022 were reconstructed piecewise. Dates have a ±1-day graphic-reading uncertainty; bars below the resolution limit may be missed. These are not raw daily gauge data.

解析：核对 GEE 与论文图近似降水，区分原始观测、图数字化和模型水量输入。固定参数替换用于检验驱动敏感性，未重新优化。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/09_Rainfall_and_data_provenance_audit/figures/P01_Daily_precipitation_4years_EN.png

95. P02_Cumulative_precipitation_4years_EN

阶段：09_Rainfall_and_data_provenance_audit

English caption：Cumulative rainfall uses each published DI sowing–harvest period, inclusive of harvest day. Open diamonds indicate the rounded seasonal precipitation inputs reported in WHCNS water balance Table 5-13, not an independently provided daily gauge series. The original AquaCrop endpoint differs by one day in 2019, 2021 and 2022; those original calendars were retained in model replays.

解析：核对 GEE 与论文图近似降水，区分原始观测、图数字化和模型水量输入。固定参数替换用于检验驱动敏感性，未重新优化。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/09_Rainfall_and_data_provenance_audit/figures/
P02_Cumulative_precipitation_4years_EN.png

96. P03_Existing_candidates_rainfall_sensitivity_EN

阶段：09_Rainfall_and_data_provenance_audit

English caption：Yield changes are relative to the original GEE forcing. Parameters, irrigation controls, initial soil water and all non-precipitation forcings are fixed. Historical models use maximum root depth 2.30 m and fMulch = 0. Treatments separate changes in total precipitation, graphic timing and intensity distribution, and synthetic redistribution of trace precipitation. These are backtests of existing candidates, not reoptimized Pareto fronts.

解析：核对 GEE 与论文图近似降水，区分原始观测、图数字化和模型水量输入。固定参数替换用于检验驱动敏感性，未重新优化。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/09_Rainfall_and_data_provenance_audit/figures/
P03_Existing_candidates_rainfall_sensitivity_EN.png

97. P04_2021_fixed_schedule_rainfall_checks_EN

阶段：09_Rainfall_and_data_provenance_audit

English caption：Each panel uses its own previously selected conditional pilot parameters at maximum root depth 1.00 m. Six evaluated proportional irrigation levels are 0, 300, 350, 400, 500 and 633.47 mm; connecting lines guide the eye and are not a new continuous optimization. Rainfall changes only; fMulch and all other inputs remain fixed within a panel. These 500-step pilot fits have not established posterior convergence. The 350-mm modeled plateau persists under the tested rainfall alternatives.

解析：核对 GEE 与论文图近似降水，区分原始观测、图数字化和模型水量输入。固定参数替换用于检验驱动敏感性，未重新优化。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/09_Rainfall_and_data_provenance_audit/figures/
P04_2021_fixed_schedule_rainfall_checks_EN.png

98. S01_Four_year_minimum_irrigation_EN

阶段：10_Single_objective_multi_algorithm_validation

English caption：GA denotes real-coded genetic algorithm, GSA generalized simulated annealing, and DE differential evolution; NSGA-II is used with one objective. Each method minimizes actual irrigation subject to a prescribed Y13 threshold. Curves show median threshold minima from its own complete sweep archive; bands span three seeds and are not confidence intervals. Open circles show actual modeled zero-irrigation yield. The star is the DI model baseline. Historical NSGA-III is an external archive with a different evaluation budget. Lines join discrete tested requirements; they do not establish global optimality.

解析：比较给定产量门槛下的最小实际灌水；算法预算、三个种子和档案规则按英文图注解释。数值一致不代表物理或田间验证。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/10_Single_objective_multi_algorithm_validation/figures/S01_Four_year_minimum_irrigation_EN.png

99. S02_2021_three_fMulch_minimum_irrigation_EN

阶段：10_Single_objective_multi_algorithm_validation

English caption：The three panels use different previously selected conditional parameter sets at fMulch 0.50, 0.75 and 0.85; this is not a parameter-fixed one-factor mulch experiment. Maximum root depth is 1.00 m. The original GEE weather, initial soil water, original event dates and four irrigation windows are retained. No terminal-storage floor is added. The 500-step parent fits have not demonstrated posterior convergence. Other plotting conventions follow S01.

解析：比较给定产量门槛下的最小实际灌水；算法预算、三个种子和档案规则按英文图注解释。数值一致不代表物理或田间验证。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/10_Single_objective_multi_algorithm_validation/figures/S02_2021_three_fMulch_minimum_irrigation_EN.png

100. S03_2021_actual_yield_irrigation_EN

阶段：10_Single_objective_multi_algorithm_validation

English caption：Markers are actual evaluated solutions, selected as the lowest-irrigation feasible candidate over three seeds for each method and requirement. The y-axis is achieved model yield, whereas S01–S02 show required yield. No missing or failed thresholds are imputed. The prior proportional response changes all original event amounts by a common factor; optimized solutions redistribute amounts among four windows. The historical NSGA-III front uses a different search budget.

解析：比较给定产量门槛下的最小实际灌水；算法预算、三个种子和档案规则按英文图注解释。数值一致不代表物理或田间验证。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/10_Single_objective_multi_algorithm_validation/figures/
S03_2021_actual_yield_irrigation_EN.png

101. S04_95pct_yield_algorithm_comparison_EN

阶段：10_Single_objective_multi_algorithm_validation

English caption：Each colored marker is the minimum feasible irrigation in one method/seed archive, including the equal 384-query continuation at this requirement. Three seeds are displayed separately. Dark ticks show historical NSGA-III archive minima, which are references rather than equal-budget competitors. The requirement is 95% of the observed yield level, not a 95% probability of success.

解析：比较给定产量门槛下的最小实际灌水；算法预算、三个种子和档案规则按英文图注解释。数值一致不代表物理或田间验证。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/10_Single_objective_multi_algorithm_validation/figures/
S04_95pct_yield_algorithm_comparison_EN.png

102. S05_2021_convergence_EN

阶段：10_Single_objective_multi_algorithm_validation

English caption：Lines show the median incumbent feasible irrigation across three seeds during the 384-query independent run; shading spans the seed minimum and maximum. Setup initial populations contain the DI baseline, ensuring a feasible anchor at this requirement. The later equally budgeted 384-query continuation is reported separately. Convergence here assesses numerical search, not correctness of the model water-stress response.

解析：比较给定产量门槛下的最小实际灌水；算法预算、三个种子和档案规则按英文图注解释。数值一致不代表物理或田间验证。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/10_Single_objective_multi_algorithm_validation/figures/S05_2021_convergence_EN.png

103. S06_Irrigation_reduction_water_budget_EN

阶段：10_Single_objective_multi_algorithm_validation

English caption：For each model, the lowest-irrigation 95%-yield candidate found across the four methods and three seeds is compared with its own DI baseline. The storage term is the complete hydrological profile at the common calendar endpoint, not the root zone alone. Other differences close the accounting for runoff, drainage, groundwater and surface-storage terms. Irrigation reductions must not be interpreted as equal reductions in ET or as validated sustainable field savings.

解析：比较给定产量门槛下的最小实际灌水；算法预算、三个种子和档案规则按英文图注解释。数值一致不代表物理或田间验证。

解释边界：须按本阶段的模型设置、参数和统计口径解释；候选不是经过独立田间验证的灌溉定额。

源表：

图像：stages/10_Single_objective_multi_algorithm_validation/figures/S06_Irrigation_reduction_water_budget_EN.png